

DERBY, Australia

WMO: 952050

Lat: 17.3706S

Lon: 123.6611E

Elev: 22

StdP: 14.68

Time Zone: 8.00 (E08)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	52.0	55.3	15.2	11.9	78.4	20.3	15.2	75.8	20.5	75.7	18.7	76.0	4.8	160	0.457

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	20.6	103.3	72.6	101.2	73.5	99.4	74.3	90.9	92.3	84.9	88.2	82.0	89.6	12.2	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
90.6	221.6	92.1	84.4	180.5	86.3	80.2	156.6	85.8	57.3	92.4	49.4	87.7	45.9	89.7	98.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.4	20.0	17.9	DB	47.2	108.9	3.1	2.2	45.0	110.5	43.3	111.8	41.5	113.0	39.3	114.6	(4)
(5)			WB	39.6	93.2	2.7	5.5	37.6	97.1	36.0	100.4	34.5	103.5	32.5	107.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	82.1	86.2	85.8	86.1	84.4	78.3	73.3	72.6	74.5	80.8	86.0	88.8	88.4	(6)
(7)		DBStd	6.78	3.55	3.42	3.10	3.41	4.24	4.58	4.12	3.80	3.78	2.94	2.58	3.41	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	4	0	0	0	0	0	2	2	0	0	0	0	0	(9)
(10)		CDD50	11703	1122	1001	1119	1032	878	698	700	760	924	1116	1163	1190	(10)
(11)		CDD65	6232	657	581	654	582	413	250	237	295	474	651	713	725	(11)
(12)		CDH74	78310	8426	7338	8350	7290	4474	2536	2581	3440	5621	8349	9912	9993	(12)
(13)		CDH80	41084	4263	3598	4278	3882	2182	987	1036	1601	3054	4667	5822	5714	(13)
(14)	Wind	WSAvg	9.0	10.2	9.4	8.0	7.4	7.9	8.2	7.8	8.2	8.7	10.0	10.7	10.9	(14)
(15)	Precipitation	PrecAvg	24.9	7.7	7.8	4.1	0.8	0.9	0.2	0.0	0.0	0.0	0.4	2.1	(15)	
(16)		PrecMax	43.4	24.3	20.1	14.6	3.1	5.5	2.6	0.5	0.0	0.4	1.6	7.6	(16)	
(17)		PrecMin	10.3	0.7	0.8	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)	
(18)		PrecStd	7.7	6.1	4.3	3.5	0.9	1.6	0.5	0.1	0.0	0.1	0.1	0.5	2.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	101.5	101.8	102.1	101.3	98.0	92.8	92.6	95.9	102.2	105.4	106.8	106.1	(19)
(20)			MCWB	78.0	77.8	74.8	73.7	67.6	67.8	64.0	66.4	64.9	68.9	69.6	73.7	(20)
(21)		2%	DB	98.7	98.5	99.4	98.5	95.2	90.1	90.3	92.7	99.0	101.8	103.1	101.8	(21)
(22)			MCWB	79.3	79.4	75.8	73.0	67.7	66.0	63.4	65.2	66.2	69.6	72.2	76.4	(22)
(23)		5%	DB	96.4	96.0	97.1	96.5	93.0	88.1	88.3	90.5	96.7	99.1	100.5	99.2	(23)
(24)			MCWB	79.8	79.4	76.9	73.9	67.3	65.3	62.9	63.8	66.8	71.3	73.6	77.4	(24)
(25)		10%	DB	94.0	93.2	94.7	94.6	90.7	86.0	86.1	88.4	94.0	96.4	98.0	96.9	(25)
(26)			MCWB	79.6	79.7	77.6	73.4	66.9	64.5	62.6	62.3	67.2	72.2	74.6	77.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	93.3	91.4	86.3	94.5	79.9	84.8	74.4	89.6	91.9	92.2	90.9	94.9	(27)
(28)			MCD B	94.4	91.7	87.6	95.7	85.3	85.1	80.0	89.9	93.0	94.0	93.0	96.6	(28)
(29)		2%	WB	83.8	83.3	82.5	83.1	76.4	75.6	70.1	75.3	79.0	80.2	82.0	84.5	(29)
(30)			MCD B	89.1	90.5	90.2	88.2	83.7	79.3	80.8	77.1	83.1	87.9	88.8	87.3	(30)
(31)		5%	WB	81.6	81.8	81.3	79.8	74.4	71.7	67.9	69.8	74.4	78.1	79.9	81.1	(31)
(32)			MCD B	90.0	90.0	89.2	87.5	83.2	80.9	80.3	79.5	82.5	87.6	89.5	90.3	(32)
(33)		10%	WB	80.6	80.8	80.4	78.4	72.5	69.1	65.8	67.7	72.6	77.0	78.8	80.1	(33)
(34)			MCD B	89.5	89.2	88.3	86.4	82.5	79.2	79.6	79.9	83.1	86.6	88.8	89.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.0	14.6	16.9	21.4	24.6	25.7	27.5	29.3	28.1	23.2	20.6	17.6	(35)
(36)			MCDBR	18.5	18.8	20.6	23.1	26.6	26.4	29.4	30.7	30.7	26.9	24.0	21.1	(36)
(37)			MCWBR	7.0	7.1	7.7	9.8	10.5	11.2	11.8	12.8	12.1	11.0	9.2	7.9	(37)
(38)		5% WB	MCD BR	16.4	16.4	17.5	19.4	20.9	21.3	23.4	25.7	25.2	21.3	19.9	18.5	(38)
(39)			MCWBR	7.7	7.4	7.5	10.2	10.7	14.5	11.3	15.1	14.7	10.4	9.6	8.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.420	0.414	0.395	0.383	0.368	0.341	0.338	0.348	0.388	0.420	0.422	0.417	(40)	
(41)		taud	2.456	2.477	2.530	2.500	2.473	2.520	2.502	2.438	2.377	2.333	2.371	2.441	(41)	
(42)		Ebn at Noon	294	292	291	283	276	279	283	290	289	287	291	294	(42)	
(43)		Edh at Noon	38	37	34	33	31	29	30	34	39	43	42	39	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2021	2035	1945	1889	1637	1547	1636	1913	2192	2341	2375	2202	(44)	
(45)		RadStd	183	217	213	78	105	84	85	55	48	74	82	137	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	+0.79	N/A	N/A	+0.97	+0.67	+0.58	N/A	N/A	+414	+416

Nomenclature: See separate page